

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2123

Treat 230 kV
October 30, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2123. The Interconnection Customer proposes a 78.9 MW battery energy storage facility interconnecting at the Treat 230 kV substation in the Ohio area. The request was received on June 22, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 648 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,494,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Hills to Nordonia 100 kV: 153.5% loading (201 MVA) following the outage of the Imperial to Chamberlin 230 kV; DFAX = 10.8%.

The Lincoln Beach to Hills 100 kV is overloaded to 141.6% of its 239 MVA emergency rating for the loss of the Imperial to Chamberlin 230 kV. The project contributes 8.3% DFAX (6.5 MW) to this overload.

Twinsburg to Chamberlin 100 kV Circuit 1: 130.0% loading (285 MVA) following the outage of the Chamberlin to Twinsburg 100 kV #2; DFAX = 5.5%.

The Chamberlin Substation to Nordonia 100 kV is overloaded to 128.9% of its 163 MVA emergency rating for the loss of the Imperial to Chamberlin 230 kV. The project contributes 13.6% DFAX (10.7 MW) to this overload.

The Chamberlin to Twinsburg 100 kV Ckt 2 is overloaded to 124.6% of its 297 MVA emergency rating for the loss of the Chamberlin Substation to Twinsburg 100 kV Ckt 1. The project contributes 5.5% DFAX (4.4 MW) to this overload.

The Portage Lakes to Weedpatch 230 kV is overloaded to 113.9% of its 1,085 MVA emergency rating for the loss of the Point Arena to West Canton 500 kV. The project contributes 5.2% DFAX (4.1 MW) to this overload.

For the loss of Trego Substation to Beaver 500 kV L, the Portage Lakes to Old Forge 230 kV loads to 103.9% of its 1,239 MVA rating. Project DFAX: 15.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hills Substation to Nordonia 100 kV L	Rebuild line with higher-capacity conductor	\$4,311,000	33% (\$1,435,000)
Lincoln Beach to Hills 100 kV	Rebuild line with higher-capacity conductor	\$3,878,000	20% (\$767,000)
Twinsburg to Chmberlin 100 kV Circuit 1	Rebuild line with higher-capacity conductor	\$2,054,000	100% (\$2,054,000)
Chamberlin Substation to Nordoia 100 kV	Rebuild line with higher-capacity conductor	\$5,576,000	54% (\$3,021,000)
Chamberlin to Twinsburg 100 kV Ckt 2	Rebuild line with higher-capacity conductor	\$2,785,000	100% (\$2,785,000)
Portage Lakes to Weedpatch 230 kV	Rebuild line with higher-capacity conductor	\$41,630,000	3% (\$1,080,000)
Old Forge to Portage Lakes 230 kV	Reconductor line	\$9,168,000	4% (\$352,000)
	Total allocated to project		\$11,494,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.